Table 1: Rayleigh–Bénard: spectral coherence RMSE per band (averaged over rollout horizons) and CRPS per rollout horizon, including the latent-space baseline LatentFM$_{\div 64}$ (autoencoder of [rozet2025]). Mean $\pm$ std over three inference seeds (42, 84, 168).

Model	Spectral coherence RMSE			CRPS		
	Low	Mid	High	01:32	33:64	65:196
WFM 1-scale	0.168 ± 0.005	$\underline{0.118 \pm 0.001}$	$\underline{0.077 \pm <0.001}$	$0.007 \pm <0.001$	$0.041 \pm <0.001$	$\underline{0.073 \pm 0.003}$
WFM 3-scale	$\mathbf{0.143 \pm 0.002}$	$\mathbf{0.109 \pm 0.001}$	$\mathbf{0.075 \pm 0.001}$	$\mathbf{0.004 \pm <0.001}$	$\mathbf{0.032 \pm <0.001}$	$\mathbf{0.069 \pm 0.001}$
FM$_{\text{pixel}}$	$\underline{0.149 \pm 0.001}$	$0.129 \pm <0.001$	0.115 ± 0.001	$\underline{0.005 \pm <0.001}$	$\underline{0.036 \pm 0.001}$	0.076 ± 0.005
FM$_{\text{fourier}}$	0.294 ± 0.002	$0.332 \pm <0.001$	$0.113 \pm <0.001$	$0.013 \pm <0.001$	0.040 ± 0.001	0.090 ± 0.001
WDNO	0.392 ± 0.001	$0.474 \pm <0.001$	$0.535 \pm <0.001$	$0.050 \pm <0.001$	$0.102 \pm <0.001$	$0.138 \pm <0.001$
LatentFM$_{\div 64}$	$0.196 \pm <0.001$	$0.159 \pm <0.001$	0.131 ± 0.001	$0.026 \pm <0.001$	$0.057 \pm <0.001$	0.088 ± 0.004

Table 2: Kinetic-energy relative MAE across rollout horizons (error normalised by the trajectory time-mean of the true kinetic energy). Mean ± std over three inference seeds (42, 84, 168).

Rollout	Turb. Rad. Layer 2D			Rayleigh–Bénard			Active Matter		
	01:32	33:64	65:97	01:32	33:64	65:196	01:16	17:32	33:77
WFM 1-scale	0.047 ± 0.001	0.113 ± 0.002	0.221 ± 0.006	0.007 ± <0.001	0.410 ± 0.005	0.379 ± 0.010	0.502 ± 0.001	0.538 ± 0.005	0.508 ± 0.001
WFM 3-scale	0.017 ± 0.001	0.067 ± 0.003	**0.147 ± 0.004**	**0.003 ± <0.001**	0.217 ± 0.006	0.348 ± 0.023	0.112 ± 0.002	0.391 ± 0.011	0.531 ± 0.003
FM_{pixel}	**0.016 ± 0.001**	**0.067 ± 0.004**	0.177 ± 0.007	0.004 ± <0.001	0.235 ± 0.005	**0.251 ± 0.010**	**0.051 ± 0.001**	**0.288 ± 0.012**	**0.320 ± 0.010**
$FM_{fourier}$	0.054 ± 0.001	0.096 ± 0.006	0.169 ± 0.011	0.015 ± <0.001	**0.157 ± 0.002**	0.919 ± 0.006	0.503 ± 0.002	0.538 ± 0.013	0.692 ± 0.008
WDNO	0.356 ± <0.001	0.225 ± 0.001	0.457 ± 0.002	1.014 ± 0.002	2.506 ± 0.007	2.309 ± 0.003	0.747 ± 0.002	0.890 ± 0.002	1.052 ± 0.002
$LatentFM_{\div 64}$	–	–	–	0.252 ± 0.003	0.574 ± 0.008	0.913 ± 0.036	–	–	–

Table 3: TRL, WFM 3-scale (haar): CRPS across rollout horizons under Gaussian perturbations of the initial conditions with standard deviation $\eta\,\sigma_c$. Mean $\pm$ std over three inference seeds (42, 84, 168).

IC noise	01:32	33:64	65:97
$\eta = 0$	0.410 ± 0.003	1.018 ± 0.011	1.178 ± 0.030
$\eta = 0.001$	0.410 ± 0.003	1.036 ± 0.016	1.180 ± 0.024
$\eta = 0.01$	0.410 ± 0.003	1.030 ± 0.016	1.190 ± 0.007
$\eta = 0.1$	0.413 ± 0.003	1.026 ± 0.040	1.203 ± 0.012

Table 4: TRL, WFM 3-scale (haar): Spectral coherence RMSE across bands under Gaussian perturbations of the initial conditions with standard deviation $\eta\,\sigma_c$. Mean $\pm$ std over three inference seeds (42, 84, 168).

IC noise	Low	Mid	High
$\eta = 0$	0.133 ± 0.001	$0.174 \pm <0.001$	$0.141 \pm <0.001$
$\eta = 0.001$	0.133 ± 0.001	0.174 ± 0.001	$0.141 \pm <0.001$
$\eta = 0.01$	0.133 ± 0.001	0.174 ± 0.001	0.141 ± 0.001
$\eta = 0.1$	$0.133 \pm <0.001$	$0.174 \pm <0.001$	0.141 ± 0.001

Table 5: Optimal mother wavelet per WFM variant and PDE system, selected by lowest spectral coherence RMSE averaged across *Low*, *Mid*, and *High* frequency bands and rollout horizons (Table **??**). In parentheses: the selected wavelet's score; the selection is made on the seed mean. Mean over three inference seeds (42, 84, 168); the standard deviation across seeds is below 5% of the mean for all entries.

	Turbulent Radiative Layer 2D	Rayleigh–Bénard	Active Matter
WFM 1-scale	haar (0.147)	db4 (0.121)	haar (0.241)
WFM 3-scale	haar (0.149)	db4 (0.109)	db2 (0.220)

Table 6: VRMSE across rollout horizons for all models and PDE systems. Deterministic baselines (WNO, FNO, TFNO) are single runs. LatentFM$_{\div 64}$ is evaluated on Rayleigh–Bénard only. Mean over three inference seeds (42, 84, 168); the standard deviation across seeds is below 5% of the mean for all entries except those marked *.

Rollout	Turb. Rad. Layer 2D			Rayleigh–Bénard			Active Matter		
	01:32	33:64	65:97	01:32	33:64	65:196	01:16	17:32	33:77
WFM-1 haar	<u>0.544</u>	0.791	<u>0.789</u>	1.782	2.020	0.788	2.847	1.541	1.079
WFM-1 db2	0.550	0.750	0.850	1.333	1.844*	0.791	5.652	2.930	1.622
WFM-1 db4	0.550	<u>0.730</u>	0.811	1.689	2.271	0.802	14.227	5.574	2.488
WFM-1 db6	**0.536**	**0.703**	**0.786**	1.441	1.949	1.025	6.732	2.927	1.553
WFM-3 haar	0.549	0.824	0.797	1.559	2.318	0.807	<u>0.848</u>	<u>1.327</u>	<u>1.022</u>
WFM-3 db2	0.655	0.893	0.981	1.051	1.729	**0.766**	0.933	1.349	1.085
WFM-3 db4	0.586	0.759	0.803	<u>1.029</u>*	1.869	<u>0.770</u>	3.242	2.271	1.418
WFM-3 db6	0.591	0.879	0.831	**0.755**	<u>1.598</u>	0.834	2.341	2.118	1.276
FM$_{\text{pixel}}$	0.574	0.891	0.830	1.319	1.962	0.821	**0.580**	1.380	1.028
FM$_{\text{fourier}}$	0.635	0.773	0.804	2.331	**1.417**	0.870	1.386	**1.126**	**1.007**
WDNO	1.517	1.834	1.953	19.262	8.171	1.392	4.475	2.181	1.400
LatentFM$_{\div 64}$	–	–	–	9.515	4.080	0.917	–	–	–
WNO	inf	inf	–	44.226	24.463	2.452	10.669	4.321	2.002
FNO	2.952	4.978	4.349	3.368	3.972	inf	1.682	3.227	1.727
TFNO	2.300	6.534	6.387	102.678	31.842	2.965	4.443	2.968	1.525

Table 7: CRPS across rollout horizons for all models and PDE systems. LatentFM$_{\div 64}$ is evaluated on Rayleigh–Bénard only. Mean over three inference seeds (42, 84, 168); the standard deviation across seeds is below 5% of the mean for all entries except those marked *.

Rollout	Turb. Rad. Layer 2D			Rayleigh–Bénard			Active Matter		
	01:32	33:64	65:97	01:32	33:64	65:196	01:16	17:32	33:77
WFM-1 haar	0.492	1.047	1.218	0.005	0.037	0.074	0.119	0.190	0.198
WFM-1 db2	0.447	1.097	1.316	0.005	0.036	0.073	0.173	0.239	0.261
WFM-1 db4	0.521	1.052	1.241	0.007	0.041	0.073	0.326	0.388	0.387
WFM-1 db6	0.451	**0.983**	1.188	0.005	0.040	0.090	0.175	0.222	0.231
WFM-3 haar	**0.410**	1.018	<u>1.178</u>	0.004	0.037	0.077	<u>0.080</u>	<u>0.169</u>	**0.191**
WFM-3 db2	0.465	1.133	1.179	**0.003**	0.032	<u>0.069</u>	0.087	0.176	0.196
WFM-3 db4	0.427	<u>0.984</u>	**1.130**	<u>0.004</u>	<u>0.032</u>	**0.069**	0.109	0.209	0.239
WFM-3 db6	0.451	1.026	1.200	0.004	**0.031**	0.079	0.100	0.197	0.213
$\mathrm{FM}_{\mathrm{pixel}}$	<u>0.418</u>	1.088	1.210	0.005	0.036	0.076*	**0.059**	**0.166**	<u>0.194</u>
$\mathrm{FM}_{\mathrm{fourier}}$	1.112	1.632	1.772	0.013	0.040	0.090	0.197	0.216	0.211
WDNO	2.460	3.374	4.284	0.050	0.102	0.138	0.218	0.245	0.266
LatentFM$_{\div 64}$	–	–	–	0.026	0.057	0.088	–	–	–

Table 8: Spectral coherence RMSE across *Low*, *Mid*, and *High* frequency bands for all models and PDE systems, averaged over rollout horizons. LatentFM$_{\div 64}$ is evaluated on Rayleigh–Bénard only. Mean over three inference seeds (42, 84, 168); the standard deviation across seeds is below 5% of the mean for all entries.

Frequency Bands	Turb. Rad. Layer 2D			Rayleigh–Bénard			Active Matter		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
WFM-1 haar	<u>0.132</u>	<u>0.175</u>	**0.134**	0.171	0.132	0.116	0.229	0.291	0.204
WFM-1 db2	0.139	0.181	0.142	0.167	0.126	0.105	0.291	0.336	0.236
WFM-1 db4	**0.132**	0.175	0.140	0.168	0.118	<u>0.077</u>	0.306	0.351	0.251
WFM-1 db6	0.133	0.175	0.141	0.196	0.131	0.093	0.283	0.346	0.242
WFM-3 haar	0.133	**0.174**	0.141	0.149	0.130	0.114	0.205	0.280	0.256
WFM-3 db2	0.139	0.188	<u>0.136</u>	**0.141**	0.113	0.079	0.199	0.262	<u>0.198</u>
WFM-3 db4	0.141	0.178	0.144	<u>0.143</u>	**0.109**	**0.075**	0.219	0.292	0.235
WFM-3 db6	0.136	0.175	0.148	0.147	<u>0.110</u>	0.080	<u>0.196</u>	**0.258**	0.260
FM$_{\text{pixel}}$	0.141	0.181	0.138	0.149	0.129	0.115	**0.183**	<u>0.258</u>	0.291
FM$_{\text{fourier}}$	0.240	0.686	0.236	0.294	0.332	0.113	0.436	0.534	**0.001**
WDNO	0.199	0.249	0.366	0.392	0.474	0.535	0.367	0.345	0.200
LatentFM$_{\div 64}$	–	–	–	0.196	0.159	0.131	–	–	–

Table 9: CRPS per t_{cool} on TRL, averaged over rollout steps. Mean over three inference seeds (42, 84, 168); the standard deviation across seeds is below 5% of the mean for all entries except those marked *.

t_{cool}	0.03	0.06	0.1	0.18	0.32	0.56	1	1.78	3.16
WFM-1 haar	1.165	1.135	0.865*	1.044	0.830*	0.798	0.806*	0.983	0.673
WFM-1 db2	1.112	1.078	0.878	1.057	0.880*	0.779	0.853*	0.976*	1.002*
WFM-1 db4	1.170	1.102	0.847*	1.084	0.875	0.849	0.879*	0.946	0.719
WFM-1 db6	1.076	1.074	<u>0.831</u>	1.041	0.823	0.773	0.741	<u>0.943</u>	<u>0.594</u>*
WFM-3 haar	1.107	1.117	0.868	<u>0.988</u>*	**0.773**	<u>0.765</u>*	**0.675**	0.961	0.595
WFM-3 db2	1.248	1.141	0.894	1.027	0.836	0.783*	0.751*	1.019	0.656
WFM-3 db4	**1.048**	**1.006**	**0.817***	**0.982**	<u>0.817</u>	**0.749**	<u>0.692</u>	**0.906**	0.632
WFM-3 db6	<u>1.075</u>	<u>1.034</u>	0.882	1.000*	0.863	0.850	0.770*	0.969	0.617
FM_{pixel}	1.220	1.126	0.875*	1.061	0.855*	0.783	0.731	0.978	**0.551***
$\text{FM}_{\text{fourier}}$	1.709	1.667	1.424	1.618	1.485	1.430	1.451	1.595	1.194
WDNO	3.330	3.292	3.183	3.419	3.369	3.409	3.459	3.552	3.424

Table 10: Spectral coherence RMSE per t_{cool} and frequency band on TRL. Mean over three inference seeds (42, 84, 168); the standard deviation across seeds is below 5% of the mean for all entries.

Model	t_{cool}	0.03	0.06	0.1	0.18	0.32	0.56	1	1.78	3.16
WFM-1 haar	Low	0.138	0.139	0.138	0.138	**0.132**	0.125	0.127	0.136	0.121
	Mid	0.190	0.194	0.189	0.189	**0.180**	0.172	0.157	0.155	0.150
	High	0.138	0.138	**0.136**	**0.140**	0.147	0.147	0.133	0.120	0.106
WFM-1 db2	Low	0.141	**0.139**	0.140	0.137	0.143	0.134	0.136	0.134	0.149
	Mid	0.192	0.195	0.189	0.193	0.189	0.179	0.170	0.163	0.164
	High	0.142	0.143	0.139	0.145	0.153	0.155	0.145	0.134	0.122
WFM-1 db4	Low	**0.136**	0.141	**0.134**	**0.134**	0.136	0.130	0.127	**0.129**	0.123
	Mid	**0.189**	**0.192**	**0.185**	**0.188**	0.184	0.177	0.162	0.154	0.148
	High	**0.136**	**0.138**	0.137	0.144	0.151	0.158	0.147	0.135	0.112
WFM-1 db6	Low	0.138	0.143	0.139	0.136	0.136	**0.124**	0.128	0.134	0.120
	Mid	0.190	0.196	0.191	0.191	0.183	0.172	0.158	0.151	0.146
	High	0.139	0.140	0.141	0.147	0.154	0.160	0.145	0.131	0.112
WFM-3 haar	Low	0.143	0.144	0.139	0.136	0.133	0.127	0.123	0.132	**0.118**
	Mid	0.194	0.199	0.188	0.190	0.181	**0.167**	**0.153**	**0.150**	**0.144**
	High	0.152	0.152	0.148	0.148	0.157	0.152	0.135	0.123	0.103
WFM-3 db2	Low	0.150	0.146	0.145	0.148	0.141	0.136	**0.123**	0.141	0.123
	Mid	0.193	0.196	0.191	0.201	0.195	0.188	0.174	0.180	0.175
	High	0.152	0.151	0.146	0.143	**0.141**	**0.140**	0.127	0.120	0.105
WFM-3 db4	Low	0.151	0.146	0.141	0.144	0.146	0.137	0.132	0.140	0.131
	Mid	0.195	0.197	0.189	0.192	0.189	0.176	0.162	0.155	0.147
	High	0.149	0.152	0.150	0.154	0.159	0.157	0.141	0.129	0.109
WFM-3 db6	Low	0.145	0.147	0.144	0.139	0.141	0.125	0.132	0.136	0.119
	Mid	0.193	0.198	0.188	0.191	0.181	0.168	0.158	0.152	0.147
	High	0.148	0.150	0.151	0.156	0.170	0.164	0.144	0.134	0.112
FM_{pixel}	Low	0.144	0.142	0.142	0.143	0.141	0.143	0.137	0.154	0.126
	Mid	0.193	0.199	0.191	0.196	0.186	0.187	0.169	0.163	0.151
	High	0.153	0.157	0.149	0.150	0.153	0.142	**0.126**	**0.113**	**0.098**
$\text{FM}_{\text{fourier}}$	Low	0.252	0.250	0.238	0.255	0.240	0.232	0.235	0.244	0.214
	Mid	0.680	0.674	0.677	0.682	0.686	0.689	0.694	0.688	0.705
	High	0.224	0.223	0.224	0.229	0.233	0.238	0.244	0.249	0.258
WDNO	Low	0.204	0.203	0.194	0.205	0.195	0.192	0.195	0.215	0.190
	Mid	0.253	0.252	0.250	0.246	0.241	0.244	0.247	0.251	0.262
	High	0.359	0.363	0.355	0.363	0.367	0.370	0.376	0.378	0.373

Table 11: CRPS per Rayleigh number Ra on Rayleigh–Bénard, averaged over Prandtl numbers. Mean over three inference seeds (42, 84, 168); the standard deviation across seeds is below 5% of the mean for all entries except those marked *.

Ra	10^6	10^8	10^{10}
WFM-1 haar	0.059	0.060*	0.051
WFM-1 db2	**0.052**	0.062*	0.052*
WFM-1 db4	0.059*	0.060	0.051
WFM-1 db6	0.060*	0.069*	0.075
WFM-3 haar	0.060*	0.060	0.055*
WFM-3 db2	0.054*	<u>0.055</u>	<u>0.049</u>
WFM-3 db4	<u>0.053</u>*	0.058	**0.048***
WFM-3 db6	0.065*	**0.055**	0.057
$\mathrm{FM}_{\mathrm{pixel}}$	0.057*	0.066*	0.051
$\mathrm{FM}_{\mathrm{fourier}}$	0.069	0.076	0.062
WDNO	0.120	0.117	0.116
$\mathrm{LatentFM}_{\div 64}$	0.087*	0.067	0.065

Table 12: CRPS per Prandtl number Pr on Rayleigh–Bénard, averaged over Rayleigh numbers. Mean over three inference seeds (42, 84, 168); the standard deviation across seeds is below 5% of the mean for all entries except those marked *.

Pr	10^{-1}	1	10^{1}
WFM-1 haar	0.091	0.050*	<u>0.029</u>
WFM-1 db2	0.088*	0.049	0.030
WFM-1 db4	0.088*	<u>0.048</u>	0.034
WFM-1 db6	0.087	0.066*	0.051
WFM-3 haar	0.092*	0.050	0.033
WFM-3 db2	<u>0.079</u>	**0.048**	0.031
WFM-3 db4	**0.079***	0.051*	**0.028**
WFM-3 db6	0.086*	0.050	0.040
$\mathrm{FM}_{\mathrm{pixel}}$	0.088*	0.054	0.033
$\mathrm{FM}_{\mathrm{fourier}}$	0.111	0.058	0.038
WDNO	0.146	0.112	0.096
$\mathrm{LatentFM}_{\div 64}$	0.101*	0.062	0.056

Table 13: Spectral coherence RMSE per frequency band on Rayleigh–Bénard vs Ra (averaged over Pr). Mean over three inference seeds (42, 84, 168); the standard deviation across seeds is below 5% of the mean for all entries except those marked *.

Model	Ra	10^6	10^8	10^{10}
WFM-1 haar	Low	0.174	0.166	0.138
	Mid	0.189	0.135	0.133
	High	0.110	0.186	0.163
WFM-1 db2	Low	0.154	0.176	0.140
	Mid	0.161	0.132	0.134
	High	0.092	0.163	0.150
WFM-1 db4	Low	0.160*	0.169	0.140
	Mid	<u>0.121</u>	0.130	0.131
	High	**0.049**	0.102	<u>0.120</u>
WFM-1 db6	Low	0.199	0.219	0.210
	Mid	0.160	0.138	0.178
	High	0.066	0.121*	0.183
WFM-3 haar	Low	0.154	0.163	0.130
	Mid	0.188	0.133	0.129
	High	0.110	0.181	0.156
WFM-3 db2	Low	<u>0.138</u>	<u>0.159</u>*	**0.127**
	Mid	0.130	<u>0.120</u>	**0.125**
	High	0.060	0.104	**0.120**
WFM-3 db4	Low	**0.137***	0.160	<u>0.130</u>
	Mid	**0.118**	**0.119**	<u>0.125</u>
	High	<u>0.052</u>	**0.097**	0.120
WFM-3 db6	Low	0.145	0.166	0.138
	Mid	0.122	0.120	0.130

Table 14: Spectral coherence RMSE per frequency band on Rayleigh–Bénard vs Pr (averaged over Ra). Mean over three inference seeds (42, 84, 168); the standard deviation across seeds is below 5% of the mean for all entries except those marked *.

Model	Pr	10^{-1}	1	10^{1}
WFM-1 haar	Low	0.187	0.144	0.147
	Mid	0.158	0.164	0.136
	High	0.192	0.155	0.111
WFM-1 db2	Low	0.180*	0.145*	0.145
	Mid	0.150	0.153	0.123
	High	0.182	0.136	0.087
WFM-1 db4	Low	0.181*	0.141	0.148
	Mid	0.138	0.133	<u>0.111</u>
	High	<u>0.120</u>	**0.087**	**0.063**
WFM-1 db6	Low	0.200*	0.210	0.217
	Mid	0.148	0.162	0.165
	High	0.129	0.150	0.091
WFM-3 haar	Low	0.171	0.139	0.137
	Mid	0.155	0.159	0.136
	High	0.189	0.150	0.108
WFM-3 db2	Low	**0.160***	0.133	**0.131**
	Mid	<u>0.128</u>	0.134	0.112
	High	0.122	0.096	<u>0.066</u>
WFM-3 db4	Low	<u>0.163</u>	**0.130**	<u>0.134</u>
	Mid	**0.126**	**0.128**	**0.108**
	High	**0.112**	<u>0.090</u>	0.066
WFM-3 db6	Low	0.172	0.137	0.139
	Mid	0.129	0.132	0.111

Table 15: Spectral Coherence RMSE averaged across *Low*, *Mid* and *High* frequency bands and rollout horizons per PDE system. For each WFM variant, **bold** denotes the optimal wavelet representation used in all subsequent experiments. LatentFM$_{\div 64}$ is evaluated on Rayleigh–Bénard only. Mean over three inference seeds (42, 84, 168); the standard deviation across seeds is below 5% of the mean for all entries.

	Turb. Rad. Layer 2D	Rayleigh–Bénard	Active Matter
WFM-1 haar	**0.147**	0.140	**0.241**
WFM-1 db2	0.154	0.132	0.288
WFM-1 db4	0.149	**0.121**	0.303
WFM-1 db6	0.150	0.140	0.290
WFM-3 haar	**0.149**	0.131	0.247
WFM-3 db2	0.154	0.111	**0.220**
WFM-3 db4	0.154	**0.109**	0.249
WFM-3 db6	0.153	0.112	0.238
$\mathrm{FM}_{\mathrm{pixel}}$	0.153	0.131	0.244
$\mathrm{FM}_{\mathrm{fourier}}$	0.387	0.246	0.324
WDNO	0.271	0.467	0.304
LatentFM$_{\div 64}$	–	0.162	–

PDE	**WFM 1-scale**	**WFM 3-scale**
TRL	1.25×	1.14×
RB	1.52×	1.38×
AM	1.46×	1.26×

Table 16: CRPS per Rayleigh number Ra on Rayleigh–Bénard, averaged over Prandtl numbers. Mean over three inference seeds (42, 84, 168); the standard deviation across seeds is below 5% of the mean for all entries except those marked *.

Ra	10^6	10^8	10^{10}
WFM-1 haar	0.059	0.060*	0.051
WFM-1 db2	**0.052**	0.062*	0.052*
WFM-1 db4	0.059*	0.060	0.051
WFM-1 db6	0.060*	0.069*	0.075
WFM-3 haar	0.060*	0.060	0.055*
WFM-3 db2	0.054*	<u>0.055</u>	<u>0.049</u>
WFM-3 db4	<u>0.053</u>*	0.058	**0.048***
WFM-3 db6	0.065*	**0.055**	0.057
$\mathrm{FM}_{\mathrm{pixel}}$	0.057*	0.066*	0.051
$\mathrm{FM}_{\mathrm{fourier}}$	0.069	0.076	0.062
WDNO	0.120	0.117	0.116
$\mathrm{LatentFM}_{\div 64}$	0.087*	0.067	0.065

Table 17: CRPS per t_{cool} on TRL, averaged over rollout steps. Mean over three inference seeds (42, 84, 168); the standard deviation across seeds is below 5% of the mean for all entries except those marked *.

t_{cool}	0.03	0.06	0.1
WFM-1 haar	1.165	1.135	0.865*
WFM-1 db2	1.112	1.078	0.878
WFM-1 db4	1.170	1.102	0.847*
WFM-1 db6	1.076	1.074	<u>0.831</u>
WFM-3 haar	1.107	1.117	0.868
WFM-3 db2	1.248	1.141	0.894
WFM-3 db4	**1.048**	**1.006**	**0.817***
WFM-3 db6	<u>1.075</u>	<u>1.034</u>	0.882
FM_{pixel}	1.220	1.126	0.875*
$\text{FM}_{\text{fourier}}$	1.709	1.667	1.424
WDNO	3.330	3.292	3.183